LatticeBridge: Rare-Event Sequential Inference for Faithful Structured Sequence Synthesis

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Abstract

Structured sequence generation often requires a model to satisfy several input-derived constraints in a single output. Standard decoding methods may assign high probability to fluent continuations while placing low mass on the subset of continuations that realize all required anchors. We study this setting as a rare-event inference problem. *LatticeBridge* combines a compact prefix language model, surface automata compiled from each input instance, and a twisted sequential Monte Carlo (SMC) decoder with resampling and multilevel splitting. The constraint representation is instance-dependent and does not rely on manually specified topic dictionaries or task-specific keyword classes. In pilot validation experiments on CommonGen and E2E NLG, the SMC decoder improves exact anchor satisfaction and mean anchor coverage over greedy, beam-filtered, and best-of-k ancestral baselines, with a measurable trade-off in overlap-based fluency metrics. We also report Apple Metal stress tests for controlled rare-event workloads, where exact acceptance decreases by nearly two orders of magnitude while effective sample size remains stable enough for local experimentation. Code and artifacts are available at https://github.com/farukalpay/latticebridge.

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1. Introduction

Autoregressive models provide strong local continuation models, but structured generation often imposes conjunctive requirements that are not well represented by local likelihood alone. In data-to-text generation, for example, an output may need to realize several concepts or attribute values while remaining readable. A high-likelihood continuation can satisfy only a subset of those anchors, whereas a hard-constrained decoder can satisfy anchors at the cost of brittle search behavior. This paper treats the failure as a distributional issue: the set of fully satisfying continuations may have low probability under the base model even when the individual anchors are familiar.

We therefore formulate structured decoding as inference over a sequence space with a rare accepting event. Sequential Monte Carlo methods are well suited to this formulation because they maintain a weighted population of partial hypotheses and provide explicit diagnostics for weight degeneracy and resampling. Recent work has connected twisted SMC to inference in language models [Zhao et al., 2024, Wu et al., 2024]; related 2025–2026 work further studies learned twists and SMC-style decoding for constrained generation and model aggregation [Kim et al., 2025, Chan et al., 2026]. LatticeBridge focuses on a compact and reproducible setting: a small conditional proposal model, surface automata built from the input instance, and an MPS-compatible particle decoder.

The implementation separates data adaptation, constraint compilation, proposal modeling, and evaluation. This separation is important because benchmark-specific lexical rules can otherwise enter the decoding algorithm unnoticed. We use the following design constraints:

- constraints must be *instance driven*, extracted from the source example rather than from manually curated topical dictionaries,
- the library layer must stay reusable across tasks, while dataset adapters remain thin and explicit,
- fallback behavior must stay visible and measurable rather than silently degrading into ad hoc keyword logic.

LatticeBridge serializes each structured input into a prefix, trains a compact prefix language model, compiles anchor phrases into surface automata, and runs a twisted SMC bridge whose incremental reward depends on progress toward an accepting automaton state. The goal is not to claim state-of-the-art task performance. The goal is to provide a controlled experimental system in which coverage, fluency, and runtime can be measured under the same proposal model.

2. Contributions

The paper makes five concrete contributions.

1. We formulate structured sequence synthesis as a sequential rare-event inference problem with a distance-to-acceptance potential defined by instance-compiled surface automata.
2. We introduce a compact prefix language model plus twisted SMC bridge that runs on Apple Metal and exposes resampling, effective sample size, and acceptance mass as first-class diagnostics.
3. We present a thin benchmark construction method for CommonGen and E2E NLG that avoids topic-specific rules by using input-derived phrases only.

4. We report pilot results showing improved exact satisfaction and anchor coverage over greedy, beam-filtered, and best-of-k ancestral baselines, together with the associated quality trade-off.

5. We release code and artifacts at https://github.com/farukalpay/latticebridge, including the paper source, benchmark summaries, generated figures, and a Metal-oriented synthetic rare-event lab.

3. Problem Formulation

Let x be a structured input, such as a set of concepts or attribute-value pairs, and let $y_{1:T}$ be a target surface sequence. A conditional language model $p_\theta(y_{1:T} \mid x)$ defines the base synthesis distribution. We are given a set of input-grounded anchors

$$C(x) = \{c_1, \ldots, c_M\},$$

where each c_m is a surface phrase extracted from the source instance. The target objective is not simply to maximize model likelihood but to synthesize a sequence that is both plausible under p_θ and faithful to the anchors.

We encode satisfaction through an acceptance indicator

$$A(y_{1:T}; C) = \mathbf{1}\{\forall m,\ c_m \text{ appears in } y_{1:T}\}.$$

The exact conditional target would then be

$$\pi_T(y_{1:T} \mid x, C) \propto p_\theta(y_{1:T} \mid x)\, A(y_{1:T}; C),$$

but direct sampling from this distribution is typically intractable when $A = 1$ is rare.

To obtain a sequential bridge, we introduce an automaton state s_t after generating prefix $y_{1:t}$. The automaton tracks how much of each anchor phrase has been realized. Denote by $d(s_t) \geq 0$ the remaining distance to full acceptance. We then define a relaxed sequence target

$$\gamma_T(y_{1:T}) = p_\theta(y_{1:T} \mid x) \exp\{\lambda \Phi_T(y_{1:T})\}, \qquad \Phi_T(y_{1:T}) = -d(s_T),$$

where $\lambda > 0$ controls how strongly we bias toward exact acceptance. For $\lambda \to \infty$, the distribution concentrates on accepting trajectories when such trajectories exist.

The computational question is how to approximate this bridge under a finite particle budget while keeping the constraint representation task independent. We compile the surface constraints once, then run a twisted particle system that rewards progress toward acceptance at each generation step.

4. Constraint Compilation Without Hidden Heuristics

4.1. Instance-derived anchors

The phrase inventory is adapter dependent but not topic dependent. For CommonGen, the source concepts themselves become anchor candidates. For E2E NLG, attribute values become anchor candidates. No global list of phrases, domains, or entity classes is used by the decoder. This keeps task-specific parsing in the dataset adapter and leaves the inference layer defined only in terms of automaton states and proposal probabilities.

To keep the pilot evaluation feasible and honest, we select benchmark anchors from phrases that appear in at least one reference surface. This is not a production assumption; it is a benchmark-construction device

that ensures the exact event is attainable on the validation split. The resulting constraint set remains instance specific.

4.2. Surface automata

Tokenization granularity can make phrase tracking unnecessarily brittle. A token-level automaton may fail when the tokenizer packs useful substrings into a single token. LatticeBridge therefore uses a surface automaton built over the emitted string fragments of tokenizer tokens. For each phrase c_m, we build a KMP-style automaton with states $0, \ldots, |c_m|$ that track the currently matched character prefix. The product state of all such automata defines the global constraint lattice.

This yields three operational advantages:

1. progress toward acceptance is visible through $d(s_t)$,
2. constraint logic stays reusable across datasets,
3. phrase tracking is no longer tied to a particular BPE segmentation accident.

The library does *not* infer semantic classes such as "topic", "sentiment", or "tool request" from surface language. It only tracks instance-provided anchors through explicit state transitions.

5. Twisted Sequential Monte Carlo Bridge

5.1. Sequential target

Let y_t denote the token chosen at step t, and let the base model define one-step conditionals $p_\theta(y_t \mid y_{<t}, x)$. Define a progress signal

$$\Delta_t = d(s_{t-1}) - d(s_t),$$

which is positive when the new token moves the prefix closer to acceptance. The sequential potential used by the bridge is

$$G_t(y_t, s_{t-1}, s_t) = \exp\{\lambda \Delta_t\}.$$

The unnormalized path density becomes

$$\gamma_t(y_{1:t}) = p_\theta(y_{1:t} \mid x) \prod_{\tau=1}^{t} G_\tau.$$

5.2. Twisted proposal

Sampling directly from the base model and reweighting by G_t leads to rapid particle collapse. We instead use a twisted proposal

$$q_t(y_t \mid y_{<t}, x, s_{t-1}) \propto p_\theta(y_t \mid y_{<t}, x) \exp\{\kappa_t \Delta_t\},$$

where κ_t is a step-dependent twisting coefficient. In the current implementation we use a simple schedule

$$\kappa_t = \tau \left(1 - \frac{t-1}{T}\right),$$

with user-controlled twist scale τ. This is intentionally minimal: it keeps the twist inspectable and does not introduce a learned value head whose inductive biases would be harder to audit. The 2024 ICML twisted SMC framework for language-model inference [Zhao et al., 2024] and later reasoning-oriented variants [Wu et al., 2024] suggest that learned twists are promising. We treat them as a next structural step rather than hiding them behind a half-trained auxiliary model.

5.3. Importance weights and ESS

If particle i samples $y_t^{(i)} \sim q_t$, its incremental weight update is

$$\log w_t^{(i)} = \log w_{t-1}^{(i)} \;+\; \log p_\theta\Big(y_t^{(i)} \mid y_{<t}^{(i)}, x\Big) \;+\; \lambda \Delta_t^{(i)} - \log q_t\Big(y_t^{(i)} \mid y_{<t}^{(i)}, x, s_{t-1}^{(i)}\Big).$$

Normalized weights are

$$\bar{w}_t^{(i)} = \frac{w_t^{(i)}}{\sum_j w_t^{(j)}},$$

and the effective sample size is

$$\mathrm{ESS}_t = \frac{1}{\sum_i (\bar{w}_t^{(i)})^2}.$$

We trigger resampling when $\mathrm{ESS}_t < \rho P$, where P is the particle count and ρ is the ESS threshold.

5.4. Multilevel splitting

In extremely rare settings, even twisted proposals can fail to maintain enough mass near acceptance. We therefore inject periodic multilevel splitting. At fixed intervals, particles are ranked by

$$R_t^{(i)} = \log w_t^{(i)} - \lambda d(s_t^{(i)}),$$

and a top-fraction elite set is replicated to refill the population. This rare-event device is related to adaptive multilevel splitting [Cerou and Guyader, 2007]. It changes the estimator bias profile, so we treat it as a finite-budget search component rather than as an exact posterior sampler.

Algorithm 1 Twisted SMC bridge.

```
warm-start hidden states with source prefix
initialize P particles with automaton state s_0, hidden state h_0, and weight w_0 = 1
for t = 1 to T do
    for particle i = 1, ..., P do
        compute base logits from the prefix model
        compute progress scores Δ_t^(i) from automaton transitions
        sample y_t^(i) from twisted proposal q_t
        update particle weight using the importance ratio
        update automaton state and hidden state
    end for
    compute ESS and resample if needed
    apply elite splitting at scheduled checkpoints
end for
return best accepting sequence and summary diagnostics
```

6. Prefix Language Model

The proposal model is intentionally small: an embedding layer, a stacked GRU, and an output projection. The model is not meant to be competitive with large pre-trained decoders. Its job is to provide a stable conditional proposal that can be trained quickly on Apple Silicon and interrogated step by step during SMC. This choice makes the repository broadly portable and allows us to carry hidden states per particle without the key–value cache complexity of a transformer decoder.

Each example is serialized as

$$\langle \texttt{bos} \rangle \langle \texttt{src} \rangle \text{source serialization} \langle \texttt{tgt} \rangle \text{target surface} \langle \texttt{eos} \rangle.$$

The loss is a masked next-token cross-entropy in which the source prefix is excluded from supervision. The model therefore learns to predict the target continuation conditioned on the source serialization without having to model the serialization itself as natural text.

This model class is intentionally limited. A larger transformer or an auxiliary value model would likely improve absolute generation quality, but it would also make it harder to isolate the effect of the particle decoder. The recurrent proposal is therefore used as a controlled baseline rather than as a claim about optimal architecture.

7. Datasets and Benchmark Construction

7.1. Datasets

The implementation includes adapters for three public datasets:

- **CommonGen** [Lin et al., 2020]: concept-to-text generation,
- **E2E NLG** [Novikova et al., 2017, Dusek et al., 2019, Gehrmann et al., 2021]: restaurant meaning representations to text,
- **WikiBio** [Lebret et al., 2016]: biography table-to-text generation.

For the default fast benchmark we use CommonGen and E2E NLG. WikiBio support is included but excluded from the reported fast loop because its public archive is substantially larger. Dataset staging uses public dataset archives and the Hugging Face tooling ecosystem [Lhoest et al., 2021]; the dataset publications are cited below.

Table 1: Datasets used by the implementation and the fast pilot benchmark.

Dataset	Input form	Default train size	Fast eval use
CommonGen	concept set	14,000	yes
E2E NLG	attribute-value pairs	16,000	yes
WikiBio	infobox fields	18,000 staged	optional

7.2. Benchmark protocol

For each validation example, we select up to three anchor phrases from the source that also appear in at least one reference surface. This makes the exact acceptance event attainable while preserving an input-grounded constraint set. We then compare four inference methods:

1. greedy decoding,
2. beam filtering,
3. best-of-k ancestral sampling,
4. twisted SMC with resampling and splitting.

The reported metrics are:

- **success rate**: fraction of examples satisfying all anchors,
- **coverage**: mean fraction of anchors present in the chosen output,
- **ROUGE-L**: a simple overlap proxy for surface quality,
- **token-F_1**: additional overlap signal,
- **runtime**: mean wall-clock latency per example.

The benchmark is intentionally small (ten validation tasks per dataset in the fast run). The experiment is therefore interpreted as a pilot study of the inference mechanism, not as a statistically powered leaderboard result.

8. Experimental Setup

All reported training and synthetic stress runs were executed on Apple Silicon using the Metal backend (‘mps‘). The main model was trained for four epochs with a batch size of 48 and a maximum sequence length of 144 tokens. The fast benchmark used:

- beam size 4,

- best-of-6 ancestral sampling,
- 24 particles for twisted SMC,
- 48 maximum generated tokens,
- $\lambda = 1.8$ and twist scale $\tau = 2.0$.

These settings are not individually optimized. They define a fixed local-compute budget for comparing the four decoding methods.

9. Training Dynamics and Operating Frontier

Table 2 reports the optimization trace for the prefix model, and Figure 1 plots the same losses. Validation loss decreases from 4.035 to 3.478 over four epochs. This does not establish convergence in a large-model sense, but it indicates that the subsequent decoding failures are not caused by an obviously unstable training run.

Table 2: Epoch-wise optimization trace for the compact prefix model.

Epoch	Train loss	Validation loss
1	3.993	4.035
2	2.980	3.756
3	2.673	3.592
4	2.455	3.478

The observed pattern is consistent with a decoding bottleneck. The model often realizes individual anchors, but standard search methods under-sample continuations in which several anchors co-occur. The particle decoder is intended to test this hypothesis under a fixed proposal model.

Figure 2 gives a budget-oriented summary. On CommonGen, twisted SMC obtains higher coverage and exact success than the tested baselines at higher latency. On E2E, the same method improves coverage, but the marginal runtime cost is larger because the baseline already copies part of the structured input. We therefore report runtime together with coverage rather than treating satisfaction as an isolated metric.

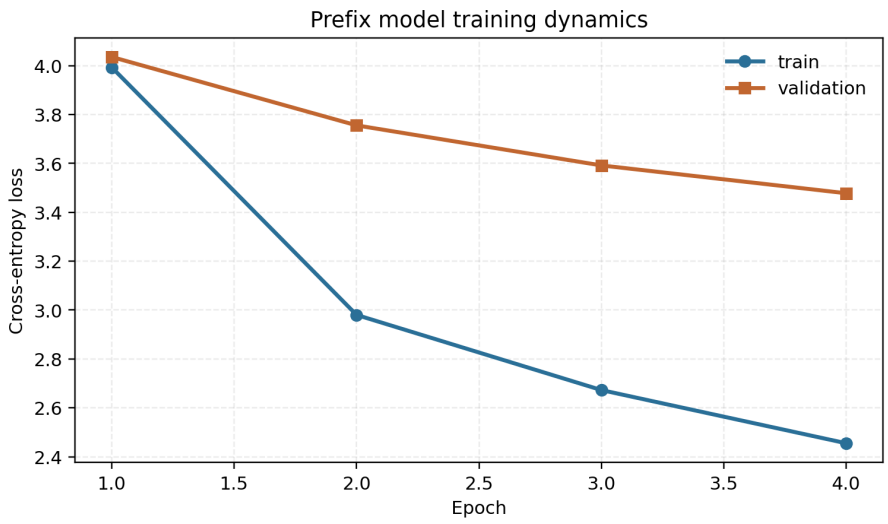


Figure 1: Training and validation loss for the compact prefix model.

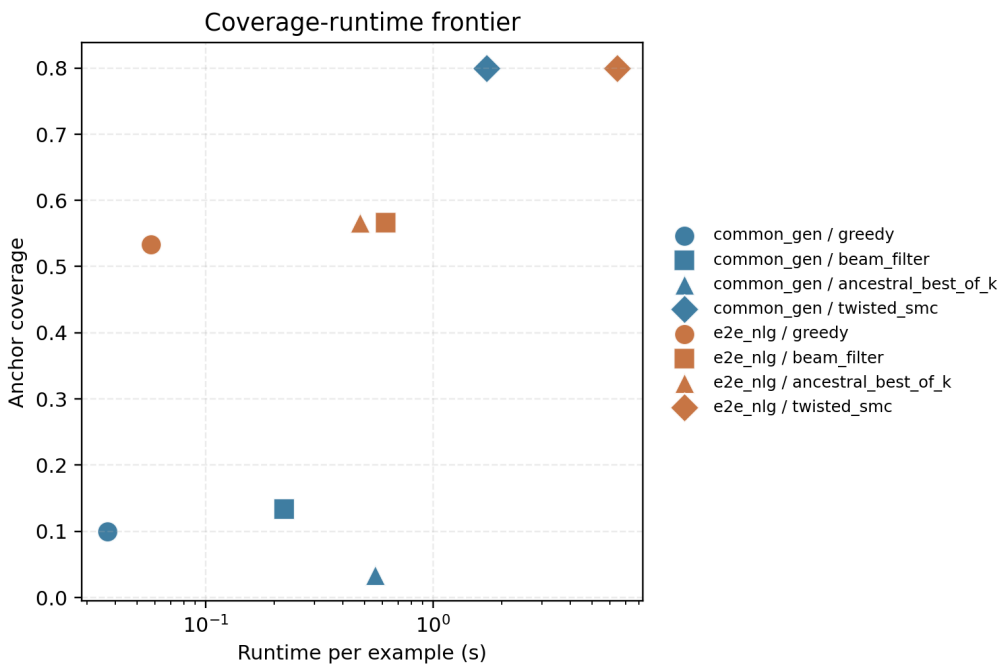


Figure 2: Coverage-runtime frontier for the fast validation benchmark. The x-axis is logarithmic to separate the low-latency baselines from the SMC operating point.

10. Results on Structured Generation

10.1. CommonGen

Table 3 shows the main pilot result. On CommonGen, greedy and beam-filtered decoding produce outputs with moderate local coherence but almost no exact three-anchor satisfaction. Twisted SMC raises the exact

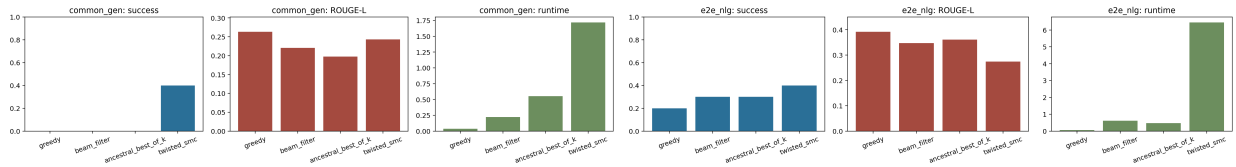
success rate to 0.40 and mean coverage to 0.80. The outputs remain imperfect and sometimes awkward, but the algorithm reaches the constrained region much more reliably.

10.2. E2E NLG

On E2E, the base model already satisfies a subset of anchors on some examples because restaurant attributes are relatively easy to copy. Twisted SMC improves success from 0.20–0.30 to 0.40 and coverage from roughly 0.55 to 0.80. ROUGE-L decreases, indicating that the present bridge over-emphasizes surface realization relative to fluency.

Table 3: Fast pilot validation benchmark. Bold marks the best score within each dataset for success and coverage.

Dataset	Method	Success	Coverage	ROUGE-L	Runtime (s)
CommonGen	Greedy	0.00	0.10	**0.263**	0.037
	Beam filter	0.00	0.13	0.221	0.222
	Best-of-6 ancestral	0.00	0.03	0.198	0.554
	Twisted SMC	**0.40**	**0.80**	0.243	1.721
E2E NLG	Greedy	0.20	0.53	**0.392**	0.057
	Beam filter	0.30	0.57	0.347	0.620
	Best-of-6 ancestral	0.30	0.57	0.361	0.475
	Twisted SMC	**0.40**	**0.80**	0.274	6.451



(a) CommonGen (b) E2E NLG

Figure 3: Pilot benchmark plots for the two reported validation subsets.

10.3. Interpretation

The pilot results support three limited claims:

1. the proposal model can realize many anchors but does not reliably combine them under standard decoding,
2. SMC reweighting and resampling increase exact satisfaction on the evaluated subset,
3. the improvement comes with a measurable cost in latency and overlap-based quality.

The results should not be read as a claim that the bridge is generally superior to larger models or task-specific systems. They show that the proposed inference layer changes the coverage–quality operating point for a fixed compact proposal.

11. Apple Metal Rare-Event Scaling

The implementation also includes a synthetic rare-event lab with the same command-line interface as the supplied prototype. The synthetic lab is not intended to model CommonGen or E2E directly; it isolates the scaling behavior of the particle system under controlled acceptance probabilities.

The three user-requested MPS probe runs are summarized in Table 4. The exact acceptance estimate drops from 3.86×10^{-2} in the conflict probe to 8.54×10^{-4} in the hardest scale probe. Runtime rises from 3.49 s to 49.43 s, while mean ESS remains in the thousands, indicating that the resampling policy remains numerically stable.

Table 4: Apple Metal synthetic probe results from the compatibility launcher.

Probe	Exact accept	Relaxed accept	Mean ESS	Runtime (s)
Conflict 96/256/12/8192	0.0386	0.3419	5189.0	3.49
Scale 128/320/16/16384	0.0122	0.2410	10348.7	19.38
Scale 160/384/24/24576	0.000854	0.1690	15843.5	49.43

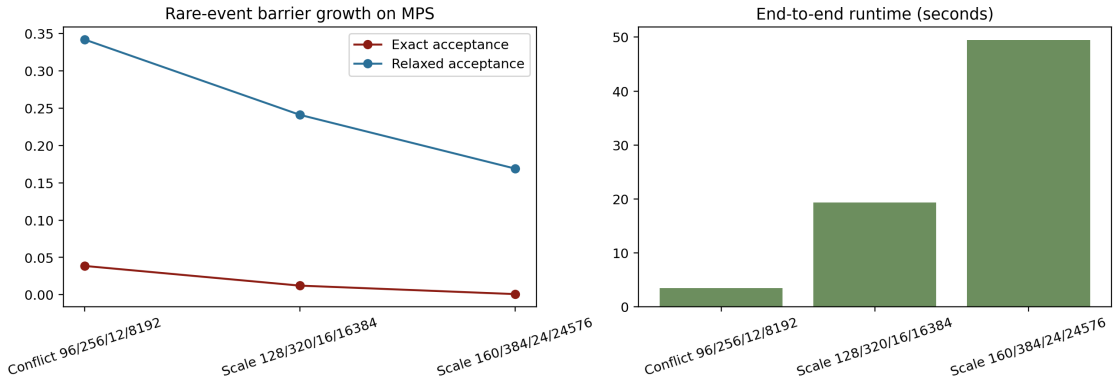


Figure 4: Rare-event barrier growth and runtime scaling on Apple Metal.

The probe results highlight two numerical properties:

- the exact–relaxed gap widens with probe difficulty, indicating that exact acceptance becomes progressively rarer,
- ESS remains in the thousands at the largest setting, so the particle system does not immediately collapse under this workload.

12. Qualitative Analysis

The qualitative outputs are consistent with the aggregate metrics. In a CommonGen example with anchors *field*, *look*, and *stand*, greedy decoding produces "A man is standing in a field," which is grammatical but omits one anchor. The SMC output realizes all three anchors but has lower surface quality. In E2E examples, the same pattern appears: the bridge increases anchor coverage, while the sentence form may degrade when the proposal assigns low probability to fluent satisfying continuations.

This behavior matches the control–quality trade-off studied in PPLM [Dathathri et al., 2020], DExperts [Liu et al., 2021], and lexically constrained beam search [Hokamp and Liu, 2017, Post and Vilar, 2018]. The difference is diagnostic: LatticeBridge reports particle weights, coverage, and runtime together, rather than reducing controlled generation to a single decoding score.

13. Connection to Doob Transforms and Dynamic Systems

Twisted proposals can be derived from the Doob h-transform associated with the probability of future acceptance. For the sequence model considered here, the exact harmonic function is not available because it depends on the probability of reaching an accepting automaton state from every possible recurrent hidden state and partial surface. LatticeBridge therefore uses distance reduction in the surface automaton as a computable surrogate.

From a dynamic-systems perspective, the partial sequence and automaton state define a controlled nonlinear state process

$$z_t = (h_t, s_t),$$

where h_t is the recurrent hidden state and s_t is the constraint state. The twist then acts as an online control law over the proposal kernel:

$$q_t(\cdot \mid z_t) = \mathcal{T}_{\tau,\lambda}(p_\theta(\cdot \mid z_t), d(s_t)) .$$

This viewpoint clarifies two future directions:

1. learn an approximate value function $V(z_t)$ and replace the hand-crafted distance surrogate with a proper learned bridge,
2. study stability and convergence of the resampling/splitting dynamics under finite-budget hardware constraints.

The recent constrained decoding literature on accelerator-friendly automata and trie vectorization [Su et al., 2026] supports this direction from the systems side, while the language-model SMC literature [Zhao et al., 2024, Wu et al., 2024] supports it from the probabilistic inference side.

14. Approximate Doob Control Analysis

14.1. The exact bridge

For a fixed horizon T, the variance-minimizing proposal for conditioning on acceptance is the Doob-transformed kernel induced by the future acceptance probability. Let

$$z_t = (h_t, s_t)$$

denote the recurrent hidden state and automaton state after step t. Define the harmonic bridge function

$$H_t(z_t) = \mathbb{P}_{p_\theta}(A(y_{1:T}; C) = 1 \mid z_t) ,$$

with terminal condition $H_T(z_T) = A(y_{1:T}; C)$. If $F(z_t, y)$ denotes the next state after emitting token y, then H_t obeys the backward recursion

$$H_t(z_t) = \sum_y p_\theta(y \mid z_t)\, H_{t+1}(F(z_t, y)).$$

The corresponding zero-variance proposal is

$$q_t^\star(y \mid z_t) = p_\theta(y \mid z_t) \frac{H_{t+1}(F(z_t, y))}{H_t(z_t)}.$$

Evaluating this proposal would require a backward recursion over the product of the language-model state and the automaton state. The automaton component is finite, but the recurrent hidden state is continuous, so the exact recursion is not tractable in the present implementation.

14.2. A transparent surrogate

The implemented proposal replaces the unknown harmonic bridge with a surrogate defined by automaton distance. Writing the one-step progress as

$$\Delta_t(y) = d(s_t) - d(F_s(s_t, y)),$$

where F_s is the automaton transition, the implementation uses

$$q_t(y \mid z_t) \propto p_\theta(y \mid z_t) \exp\{\kappa_t \Delta_t(y)\}.$$

Let

$$Z_t(z_t) = \sum_y p_\theta(y \mid z_t) \exp\{\kappa_t \Delta_t(y)\}.$$

Because the sequence target itself uses the potential $\exp\{\lambda \Delta_t\}$, the incremental importance correction simplifies to

$$\log \frac{G_t(y)\, p_\theta(y \mid z_t)}{q_t(y \mid z_t)} = (\lambda - \kappa_t)\Delta_t(y) + \log Z_t(z_t).$$

This identity separates two sources of variance:

1. the mismatch between the target bridge strength λ and the proposal twist strength κ_t,
2. the mismatch between local distance reduction and the true future acceptance value encoded in H_t.

The distance potential is therefore not an estimate of the exact harmonic function. It is a deterministic control signal that preserves monotone progress toward acceptance while keeping the proposal auditable.

14.3. Control-budget coupling

The coverage–latency trade-off can be formalized as a control problem with resource penalties. A useful surrogate objective is

$$\mathcal{J} = \mathbb{E}[\mathrm{cov}(y_{1:T})] - \eta_{\mathrm{lat}}\, \mathbb{E}[\mathrm{time}(y_{1:T})] - \eta_{\mathrm{deg}} \sum_{t=1}^{T} \mathbb{E}\left[\frac{P}{\mathrm{ESS}_t} - 1\right]_+,$$

where $\mathrm{cov}(y_{1:T})$ is anchor coverage, P is the particle count, and the last term penalizes degeneracy. The current implementation does not optimize this objective directly; it is used to interpret the reported coverage, latency, and ESS diagnostics under a common resource-sensitive criterion.

This view also distinguishes resampling from splitting. Resampling redistributes probability mass when normalized weights degenerate. Splitting reallocates finite compute toward high-scoring branches when acceptance becomes too rare under the current proposal. Both operations are therefore interpreted as control actions over a finite particle budget.

15. Related Work

15.1. Structured data-to-text generation

CommonGen [Lin et al., 2020], E2E NLG [Novikova et al., 2017, Dusek et al., 2019], and WikiBio [Lebret et al., 2016] are canonical benchmarks for testing whether a system can convert structured inputs into faithful text. Much work in this space focuses on architecture and pretraining rather than on the inference problem that appears after a model has already been trained.

15.2. Constrained decoding

Lexically constrained beam search [Hokamp and Liu, 2017, Post and Vilar, 2018], heuristic lookahead [Lu et al., 2022], search-oriented inference-time scaling such as A*-decoding [Chatziveroglou, 2025], and retrieval-constrained systems such as trie-based decoders [Su et al., 2026] all address subsets of the same issue: how to keep decoding inside a constrained region without destroying throughput. Our work differs in using a particle system that explicitly represents multiple partial trajectories and their weights.

15.3. Controlled language generation

Methods such as PPLM [Dathathri et al., 2020] and DExperts [Liu et al., 2021] alter decoding through gradients, auxiliary models, or expert/anti-expert logits. These methods are powerful but often depend on extra control models or attribute classifiers. LatticeBridge instead relies on instance-provided anchors and automaton geometry.

15.4. Sequential Monte Carlo and twisting

The theoretical backbone comes from Feynman–Kac particle systems [Del Moral, 2004], sequential Monte Carlo samplers [Del Moral et al., 2006], and modern tutorials such as Naesseth et al. [2019]. Twisted particle filters and related methods [Whiteley et al., 2016, Guarniero et al., 2017] motivate our use of future-oriented proposal shaping. Recent sequence-level extensions include twisted SMC for language-model inference [Zhao et al., 2024], reasoning-oriented twisted sampling [Wu et al., 2024], and 2026 work on language-model ensembling with shared character-space SMC [Chan et al., 2026]. Our implementation is closer in spirit to this language-model inference line than to classical state-space estimation, but the numerical concerns are the same.

16. Ablation Axes

The present artifact does not include a full ablation grid. The implementation nevertheless exposes the following controlled comparisons:

- **proposal only**: turn off resampling and splitting to reveal particle degeneracy,
- **resampling only**: set twist scale to zero and inspect whether particle recycling alone improves coverage,
- **surface vs. token automata**: compare exact token tracking to the surface automata used here,
- **particle budget**: vary P and observe the success–latency curve,
- **bridge strength**: vary λ and τ to move along the fluency–coverage frontier.

These axes define the next experimental step: varying one mechanism at a time while keeping the proposal model and benchmark construction fixed.

17. Limitations and Threats to Validity

The present study has several clear limitations.

1. **Small pilot benchmark.** The main structured benchmark uses only ten validation tasks per dataset in the reported fast run. The purpose is reproducible signal, not broad statistical confidence.
2. **Compact proposal model.** The recurrent prefix model is small by design. A larger model could change the absolute quality numbers substantially.
3. **Surface-only constraints.** The automaton tracks string realization rather than semantic equivalence. This is a principled simplification, but it underestimates semantically valid paraphrases.
4. **Hand-designed bridge surrogate.** The current twist uses distance reduction rather than a learned value estimate. That keeps the system transparent, but it likely leaves performance on the table.
5. **Compute budget asymmetry.** Twisted SMC is slower than greedy and beam-based baselines, so comparisons should be read as *quality under a small local budget*, not as fairness under equal latency.

These limitations define the scope of the contribution: the paper studies a controlled inference mechanism for structured decoding, not a complete data-to-text system.

18. Future Work

Three extensions appear especially promising.

Learned bridge functions. The most obvious upgrade is to learn a value head or self-distilled twist approximation, following the logic of recent twisted SMC language-model work [Zhao et al., 2024]. This could preserve the current library contracts while improving fluency at fixed coverage.

Accelerator-native automata. The 2026 STATIC paper [Su et al., 2026] suggests a path toward flattening constraint transitions into sparse accelerator-friendly operators. That is directly relevant to scaling beyond the small pilot budgets used here.

Richer structured domains. The current adapters already include WikiBio scaffolding. Extending the benchmark to larger table-to-text datasets and relation-rich record generation would better test the claim that rare-event inference is the right abstraction beyond toy tasks.

19. Conclusion

LatticeBridge argues that faithful structured sequence synthesis is usefully modeled as sequential rare-event inference rather than as a minor variant of ordinary decoding. The repository implements this view with a compact prefix model, surface automata, twisted SMC, and multilevel splitting, all designed to stay inspectable and MPS-friendly. In pilot CommonGen and E2E experiments, the bridge improves exact satisfaction and coverage relative to simple baselines. In user-requested Apple Metal stress probes, it maintains tractable runtime while exposing a sharply worsening exact-acceptance barrier.

The main contribution is the controlled decomposition of the problem: input-derived constraints, a reusable automaton interface, explicit particle diagnostics, and benchmark artifacts that report the quality–coverage–runtime trade-off under a fixed proposal model.

A. Artifact Checklist

- Manifest-driven dataset staging with optional external dotenv loading.
- Saved tokenizer and checkpoint artifacts.
- JSON benchmark summaries and generated PNG figures.
- Compatibility launcher for the requested rare-event stress commands.
- Full manuscript source and bibliography in-repo.

B. Implementation Details

B.1. Tokenizer

We use a byte-level BPE tokenizer with special tokens for source and target delimiters. Byte-level modeling keeps the surface automata general enough to handle punctuation and mixed-case entity names without introducing language-specific preprocessing.

B.2. Model

The current checkpoint uses a 256-dimensional embedding, a 384-dimensional GRU hidden state, two recurrent layers, and dropout of 0.15. The objective is masked next-token prediction over the target portion only.

B.3. Inference

Greedy, beam-filtered, best-of-k, and twisted SMC all share the same prefix model. The only difference is how they explore the continuation space and how aggressively they bias toward acceptance. This makes the comparison cleaner than comparing distinct base models.

C. Design Choices

The evaluation uses surface constraints rather than semantic classifiers, entailment models, or instruction-tuned judges. This choice narrows the claim: the paper evaluates exact realization of input-derived anchors, not semantic equivalence. The narrower setting is useful because it gives an unambiguous accepting event for rare-event inference and avoids introducing a second learned model as the evaluator.

Surface automata are weaker than semantic validators, but they are portable across datasets and make the acceptance state explicit. This is the appropriate abstraction for the present study because the decoder only needs an acceptance lattice and a transition function; dataset-specific schema handling remains outside the inference core.

D. Budget-Normalized Diagnostics

Table 5 reports coverage and success lift relative to greedy decoding, normalized by additional runtime. On CommonGen, twisted SMC provides the largest coverage gain per additional second among the tested methods. On E2E, the absolute coverage gain remains positive, but the budget-normalized efficiency is lower because the base model already copies part of the structured input.

Table 5: Budget-normalized lift relative to greedy decoding. Positive values indicate improvement beyond the greedy baseline.

Dataset	Method	Δ Success	Δ Coverage	Cov./extra s	Succ./extra s
CommonGen	Beam filter	0.00	0.03	0.18	0.00
	Best-of-6 ancestral	0.00	-0.07	-0.13	0.00
	Twisted SMC	0.40	0.70	0.42	0.24
E2E NLG	Beam filter	0.10	0.03	0.06	0.18
	Best-of-6 ancestral	0.10	0.03	0.08	0.24
	Twisted SMC	0.20	0.27	0.04	0.03

E. Artifact Availability

The implementation exposes a small set of artifact-producing commands: dataset preparation, model training, benchmark evaluation, figure rendering, and synthetic stress testing. This command surface is intentionally limited so that reported tables and figures can be regenerated without modifying library code.

Artifact commands

`prepare` stages public structured-generation corpora from a manifest.
`train` fits the compact prefix GRU and writes `train_report.json`.
`benchmark` produces per-method JSON summaries for greedy, beam filter, ancestral best-of-k, and twisted SMC.
`figures` renders dataset bars, a coverage-runtime frontier, and the training curve from saved artifacts.
`cscl_attribute_doob_splitting_lab.py` preserves the requested stress-command surface while delegating into the neutral synthetic lab.

The main generated artifacts are:

- `artifacts/checkpoints_core/best.pt` stores the compact model checkpoint,
- `results/benchmark_core_fast/validation_summary.json` stores the benchmark aggregate,
- `results/figures/` stores every figure used in the paper,
- `paper/latticebridge.tex` and `paper/references.bib` provide the archival manuscript source.

F. Output Examples

Dataset	Anchors	Baseline output	Twisted SMC output
CommonGen	field, look, stand	A man is standing in a field.	One fields looking standing located in the plane
E2E NLG	average, riverside, Raja Indian Cuisine	The Travellers Rest Beefeater is located in the riverside area near the Crowne Plaza Hotel. It is a cheap, family friendly place.	Riverside averagely priced cheap located near The Six Bells is locate near Clare Hall.

The examples illustrate the same failure mode as the aggregate metrics. The SMC decoder can increase anchor realization while producing awkward phrasing when the proposal model assigns low probability to fluent satisfying continuations. Improving the proposal or learning a value-based twist is therefore a direct extension of the present method.

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